Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 3a - Decision Making as Supervised Learning (part 1)

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So far: We explored various approaches to decision-making.

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Today: focus on model-free supervised learning appraches.

Supervised Learning

Learning from Data

A central goal of **machine learning** is to automatically discover patterns from data. Instead of manually programming rules, we provide the system with data and allow it to learn a model.

Examples of tasks that can be learned from data:

- Recognizing objects in images
- Predicting house prices
- Detecting anomalies in sensor readings
- Classifying medical images



Different learning paradigms exist depending on the type of data and feedback available.

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Supervised Learning

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Supervised learning is a paradigm in machine learning where a model is trained using labeled examples.

It is one of the most common learning paradigms (and the core of most foundation models).

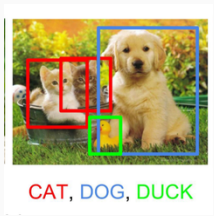
Supervised learning answers the question:

Assuming an unknown true distribution $\mathcal{D}$, how can we learn to make predictions from labeled examples sampled from $\mathcal{D}$?

Supervised Learning Examples

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- Recognizing objects in images **using images in which objects were identified**.
- Predicting house prices **based on examples of house prices under certain conditions**
- Detecting anomalies in sensor readings **based on reading for which the actual values are known**
- Classifying medical images **based on labels provided by human doctors**

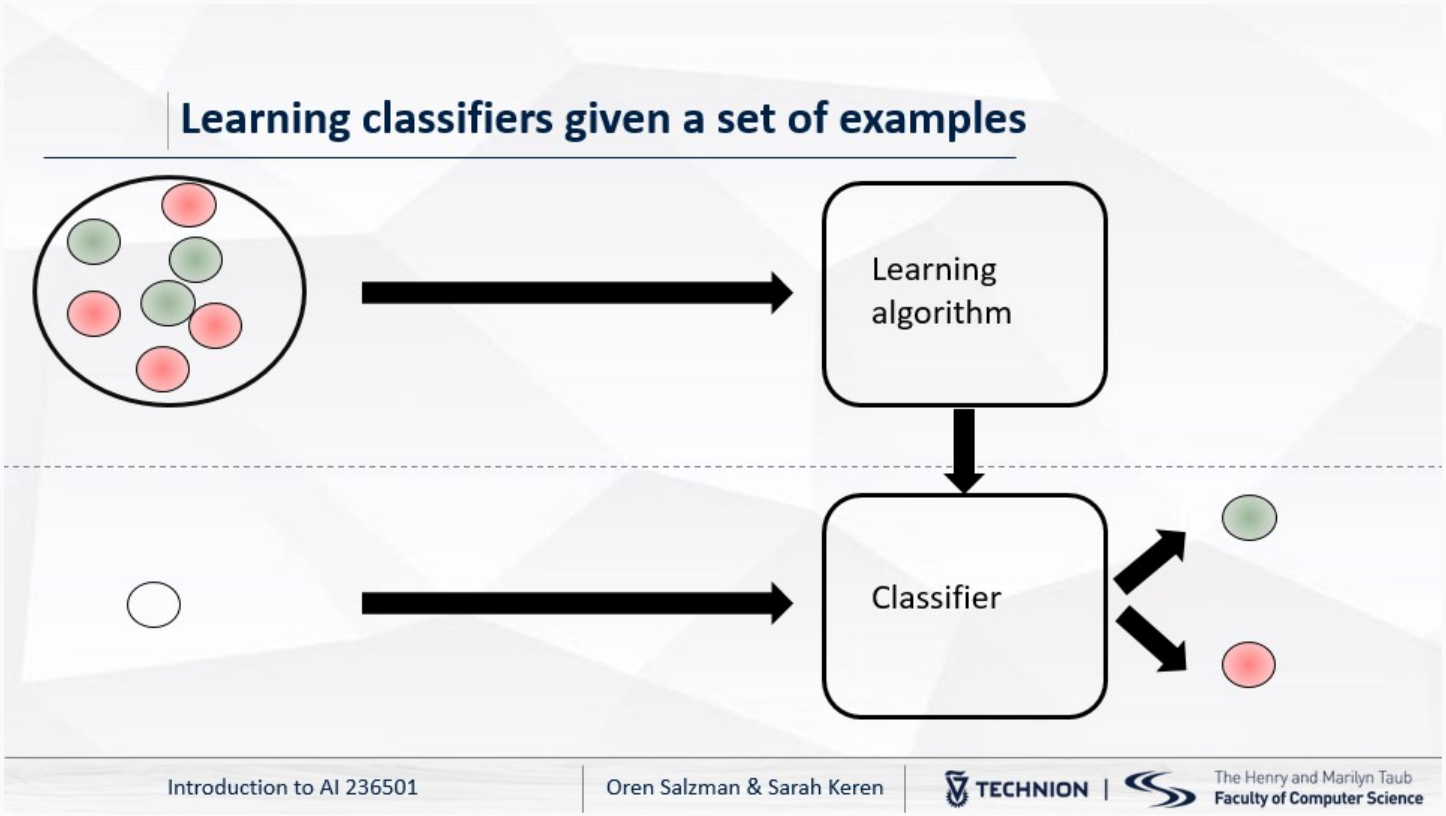








Supervised Learning in IAI



In IAI, we used supervised learning to learn the underlying model of the environment.
How is this related to computing policies and to decision-making?

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Algorithmic Setup

Supervised Learning Components

A distribution $\mathcal{D}$, from which samples are drawn i.i.d. (independent and identically distributed)

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Supervised Learning Components

A distribution $\mathcal{D}$, from which samples are drawn i.i.d. (independent and identically distributed)

A sample, or labeled example (x_i, y_i) is a pair consisting of:

- an **input** (feature vector) x_i
- the corresponding label or target value y_i

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A **sample set**: $S = \{(x_i, y_i)\}_{i=1}^{m}$ is used to learn a **predictor** $h(x) \in H$ from a **hypothesis class** H that generalizes well with respect to the underlying distribution $\mathcal{D}$, i.e., predicts the correct label for new inputs $(x, y) \sim \mathcal{D}$ with high probability.

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Supervised Learning Components

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When the label y takes values in a **discrete set** (e.g., $\{0, 1\}$ or a set of classes), the task is called **classification**.

When the label y takes values in a **continuous space** (e.g., $y \in \mathbb{R}$), the task is called **regression**.

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Supervised Learning Setup: Example

A sample includes information x_i about a person (age, gender, city of birth, ...) and her height y_i.

$x_1 = \{35, F, NY\}, \quad y_1 = 1.58$

$x_2 = \{5, M, TLV\}, \quad y_2 = 0.7$

We seek a predictor $h(x) \in H$ among a hypothesis class H, e.g., linear functions, decision trees, neural networks.

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We seek a predictor $h(x) \in H$ among a hypothesis class H, e.g., linear functions, decision trees, neural networks. For example, we may seek a linear model

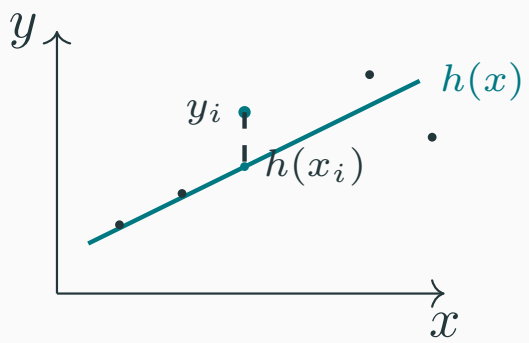
$$H = \left\{ h_w(x) = w^\top x + b \;\middle|\; w \in \mathbb{R}^d,\ b \in \mathbb{R} \right\}$$

where w are the weights and b is the bias.

Loss and Risk

In supervised learning, we aim to **learn a hypothesis that minimizes prediction error on unseen data**.

Loss $\ell(h(x), y)$: point-wise error on a **single example**



Expected Risk $\mathcal{R}(h)$: expected loss over the **data distribution**

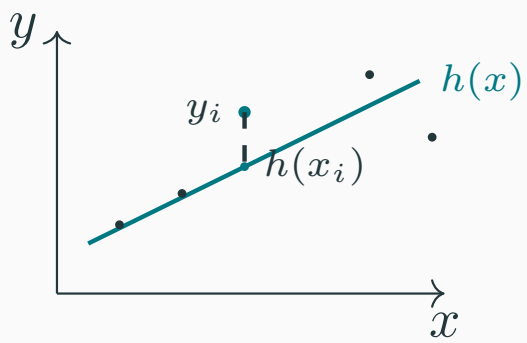
$$\mathcal{R}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$$

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Loss and Risk

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Loss $\ell(h(x), y)$: point-wise error on a **single example**



Expected Risk $\mathcal{R}(h)$: expected loss over the **data distribution**

$$\mathcal{R}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$$

Since we typically do not have access to the true distribution $\mathcal{D}$, we rely instead on a **finite dataset** $(x, y) \sim \mathcal{D}$ to compute the **Empirical Risk** $\hat{\mathcal{R}}(h)$, i.e., the average loss over the **training data**.

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Empirical Risk Minimization

Given a sample set $S = \{(x_i, y_i)\}_{i=1}^m$ and a hypothesis class H, we evaluate each hypothesis by its average loss on the training data.

For a hypothesis $h \in H$, the **empirical risk** is

$$\hat{\mathcal{R}}(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(h(x_i), y_i)$$

The learning problem is to find a hypothesis with minimal empirical risk:

$$\hat{h} = \arg\min_{h \in H} \hat{\mathcal{R}}(h) = \arg\min_{h \in H} \frac{1}{m} \sum_{i=1}^{m} \ell(h \in h(x_i), y_i)$$

Interpretation: choose the predictor whose predictions incur the smallest average loss on the observed examples.

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Examples of Loss Functions

Classification

- 0-1 Loss

$$\ell(h(x), y) = \begin{cases} 0 & h(x) = y \\ 1 & h(x) \neq y \end{cases}$$

- Cross-Entropy Loss (used with probabilistic classifiers)

$$\ell(h(x), y) = -\log P(y|x)$$

Regression

- Squared Loss (Mean Squared Error)

$$\ell(h(x), y) = (h(x) - y)^2$$

- Absolute Loss

$$\ell(h(x), y) = |h(x) - y|$$

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Supervised Learning Setup

Input: A finite data set generated from an unknown distribution $(x, y) \sim \mathcal{D}$ over inputs and labels, and a hypothesis class H

The data set is assumed to be sampled **independently and identically distributed** from some underlying distribution $\mathcal{D}$.

Optimality: a hypothesis that minimizes the expected loss:

$$h^\star = \arg\min_{h \in H} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\ell(h(x), y)\right]$$

Empirical Objective: Since the true distribution $\mathcal{D}$ is unknown, we approximate this objective using the empirical loss:

$$\hat{h} = \arg\min_{h \in H} \frac{1}{m} \sum_{i=1}^{m} \ell(h(x_i), y_i)$$

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Supervised Learning Setup

In practice, we split the available dataset into three parts:

Training set S_{train}:

- used to **learn the model parameters**
- optimize empirical risk:

Validation (evaluation) set S_{val}:

- used to **tune hyperparameters** (e.g., model size)
- helps detect **overfitting**

Test set S_{test}:

- used **only once at the end**
- provides an **unbiased estimate of generalization performance**

Never use test data during training or model selection.

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Supervised Learning Algorithms

There are several approaches to finding (or approximating) $\hat{h}$:

- **Analytical solutions** Closed-form minimization (e.g., linear regression with squared loss)
- **Iterative optimization** Gradient-based methods (e.g., SGD) for optimizing differentiable models, and search-based or gradient-free methods for non-differentiable models
- **Structured / symbolic methods** Search over hypothesis space (e.g., decision trees, rule learning)
- **Kernel / instance-based methods** Implicit hypothesis spaces (e.g., SVMs, nearest neighbors)
- **Pretrained / foundation models** Start from a large pretrained model and adapt:
  - fine-tuning on S
  - prompting / in-context learning

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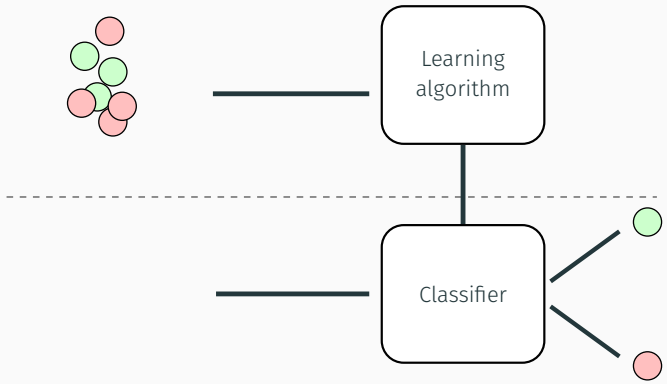
Example Approach: Iterative Supervised Learning via ERM

Iterative Supervised Learning via ERM

Require: Training set $S = \{(x_i, y_i)\}_{i=1}^{m}$, hypothesis class H, loss ℓ

Ensure: Learned hypothesis $\hat{h}$

1: Initialize hypothesis $h \in H$
2: **repeat**
3: Update h using optimization (e.g., gradient descent)
4: **until** convergence
5: $\hat{h} \leftarrow h$
6: **Return** $\hat{h}$



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Example Update Rule: Gradient Descent

A common choice for the update function is **gradient descent**, which improves the current hypothesis by moving it in the direction that reduces the empirical risk (loss).

If we let the empirical risk be $\hat{\mathcal{R}}(h) = \frac{1}{m}\sum_{i=1}^{m} \ell(h(x_i), y_i)$, each iteration becomes:

$$h \leftarrow h - \eta \nabla \left(\frac{1}{m} \sum_{i=1}^{m} \ell\big(h(x_i), y_i\big) \right)$$

where:

- $\eta > 0$ is the **learning rate**
- $\nabla \hat{\mathcal{R}}(h)$ is the gradient of the empirical risk with respect to the model parameters

Gradient descent repeatedly adjusts the model parameters to reduce the training error.

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Example Update Rule: Gradient Descent

In modern methods, deep learning is often used.
We parameterize the hypothesis as $h_\theta \in H$ with parameters θ.
The empirical risk is: $\hat{\mathcal{R}}(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell\big(h_\theta(x_i), y_i\big)$ and each iteration is:

$$\theta \leftarrow \theta - \eta \nabla_\theta \left(\frac{1}{m} \sum_{i=1}^{m} \ell\big(h_\theta(x_i), y_i\big) \right)$$

.

We iteratively update the parameters to reduce the empirical risk until convergence.

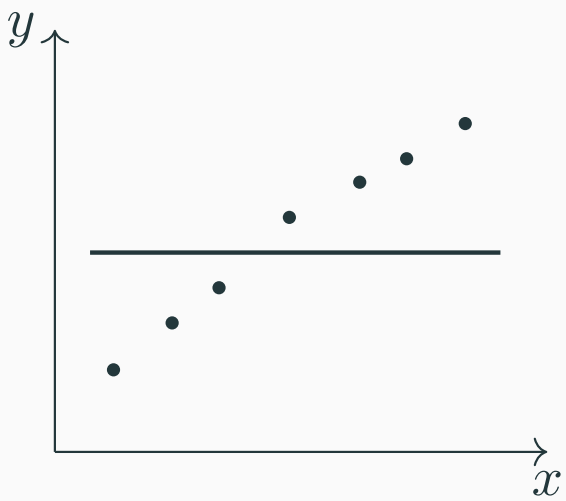
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Challenges

Supervised Learning : Key challenges

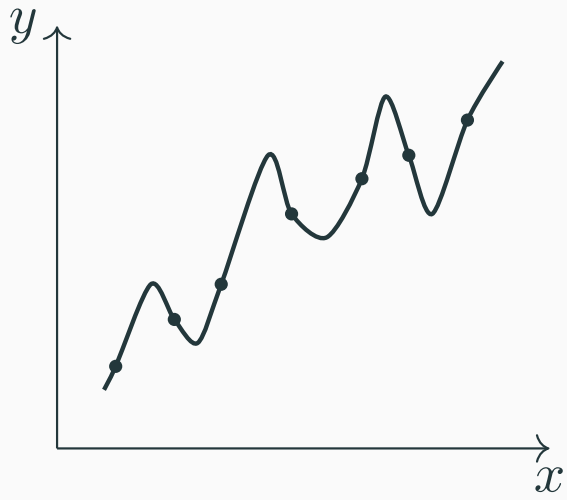
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Underfitting



Avoid **underfitting**: using a model that is too simple to capture patterns.

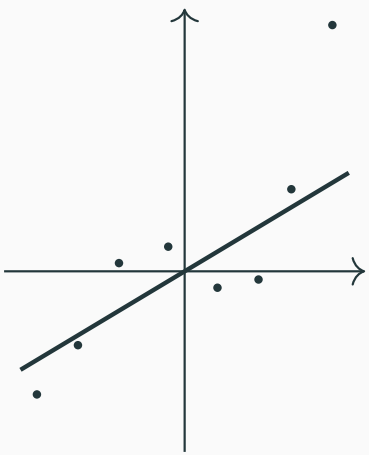
Overfitting



Avoid **overfitting**: fitting the training data too closely while failing to generalize.

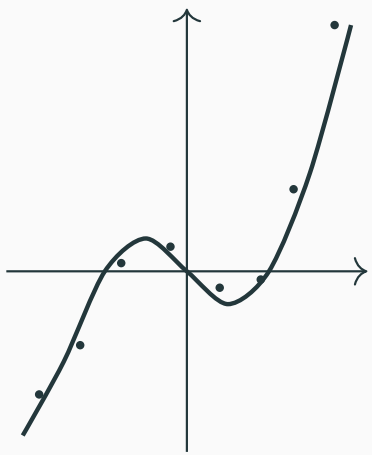
Example: Choosing Model Complexity

Assume the data is generated from an unknown cubic relationship: $y = x^3 - x + \epsilon$ where ϵ is a small noise term.



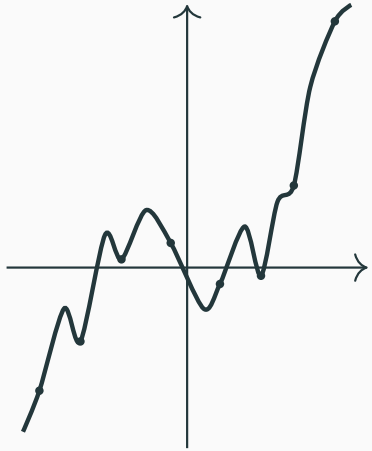
Fit with a **linear model**:

Too simple to capture the cubic trend.



Fit with a **degree-3 polynomial**:

Captures the underlying structure well.



Fit with a **n-degree polynomial**:

Fits the noise rather than the true pattern.

The goal is to find a model complexity that **generalizes well**.

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Finding an Effective Loss Function

A sample includes information x_i about a person (age, gender, city of birth, ...) and her height y_i.

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$x_2 = \{5, M, TLV\}, \quad y_2 = 0.7$

What is a helpful loss measure?

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Finding an Effective Loss Function

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What is a helpful loss measure?

Finding an Effective Loss Function

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For some prediction tasks, there is no obvious notion of **distance between outputs**.

Possible choices encode different assumptions:

- **0–1 loss**: treat all incorrect predictions equally.
- **Cost matrix**: specify a penalty for each type of mistake.
- **Representation distance**: compare output embeddings, assuming their geometry reflects relevant similarity.

Challenge: How to define a loss that captures meaningful errors when the output space provides no natural distance?

Finding an Effective Loss Function

The loss $\ell(h(x), y)$ specifies how undesirable a prediction is.

Choosing it is part of **defining the learning problem**.

- **Meaning of the output:** Numerical differences are meaningful for heights, but not for arbitrary class identifiers.
- **Severity of errors:** Should a large error receive a proportionally larger penalty, or a much larger one?
- **Consequences of errors:** Are underestimation and overestimation equally costly?

A loss can express a notion of "distance,"
but need not be a mathematical metric.

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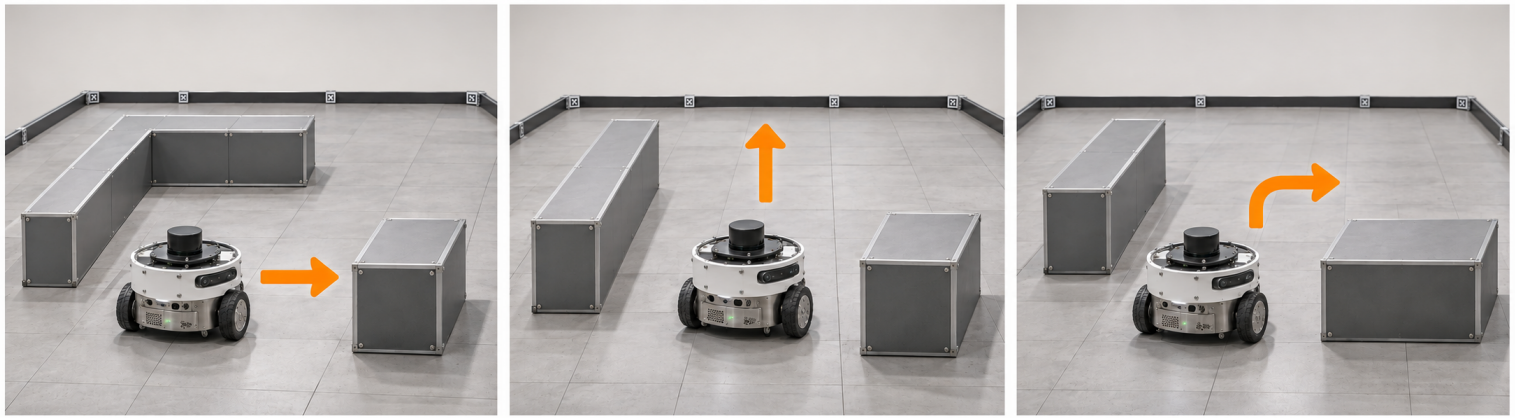
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From Supervised Learning to Decision-Making

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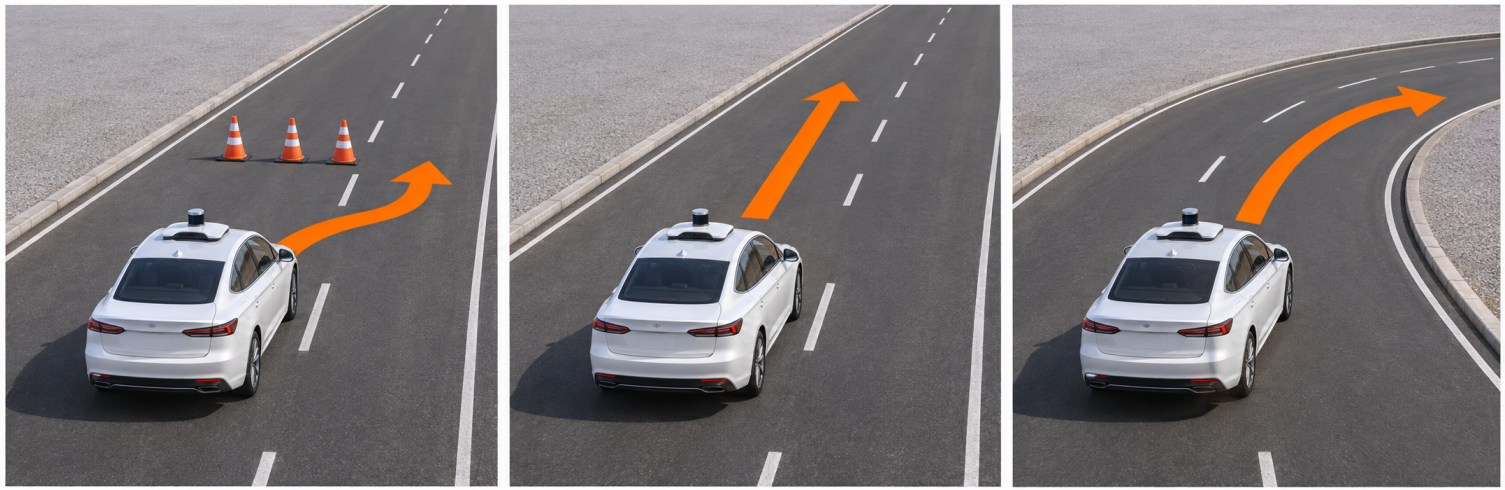
How does supervised learning relate to decision-making?

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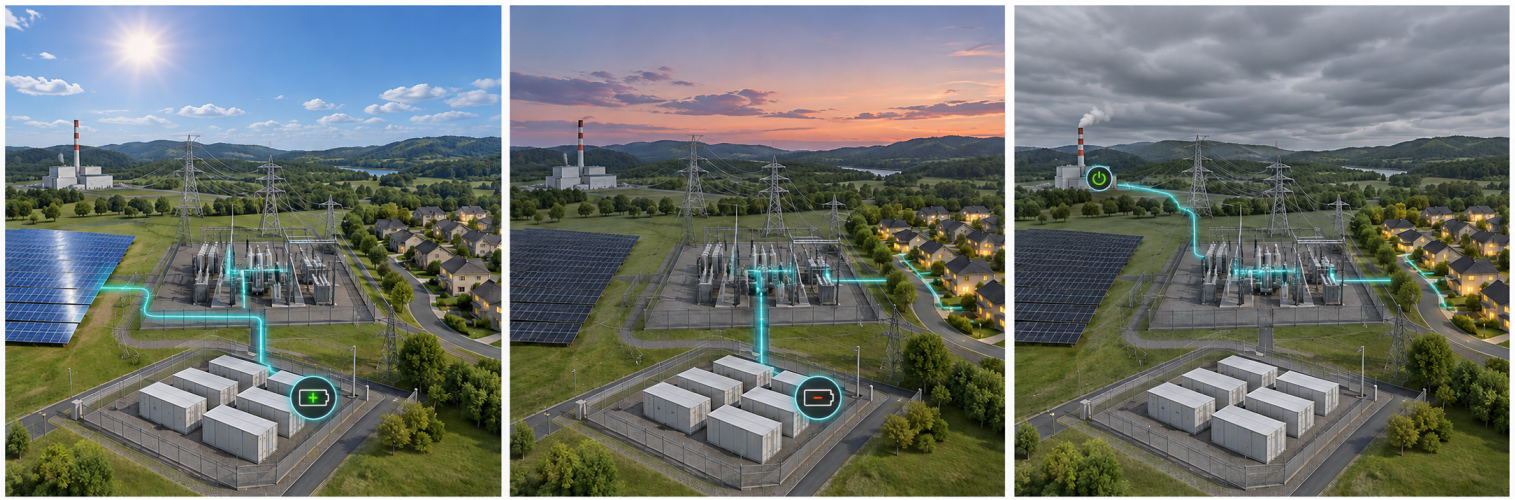
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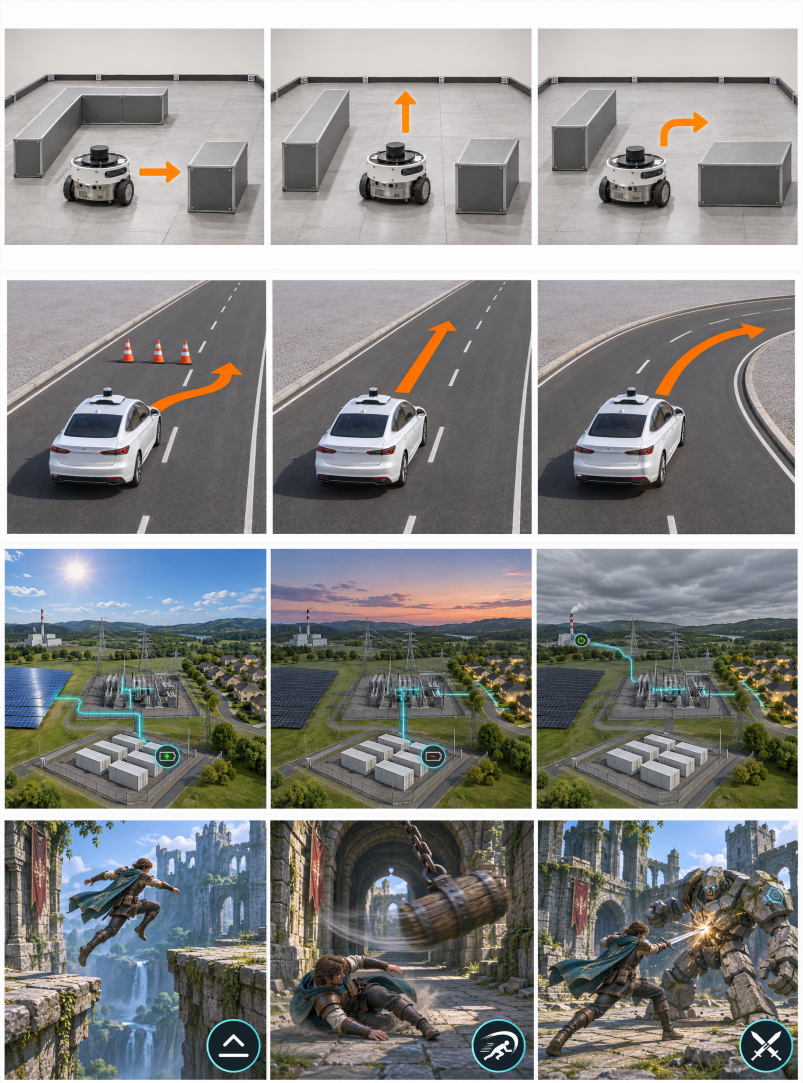
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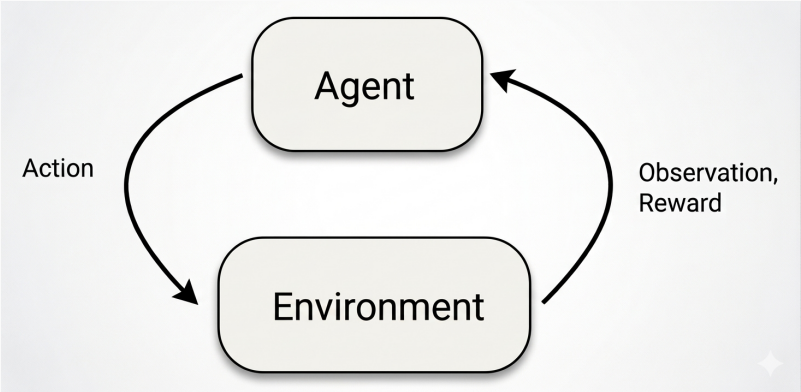
Challenges in supervised learning to decision-making?



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Supervised learning relies on the assumption that there is a given dataset and that data is **i.i.d.** from some underlying distribution $\mathcal{D}$.



Decision-making involves **interaction** with the environment.

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Challenge: Data is no longer i.i.d.

During interaction, observations are **temporally dependent**, and the agent's actions influence the data it encounters:

$$a_t \sim \pi(\cdot \mid s_t), \qquad s_{t+1} \sim P(\cdot \mid s_t, a_t).$$

Why does this matter?

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Why does this matter?

- Consecutive samples are correlated, so additional samples may provide less new information than independent samples.
- Changing the policy changes the distribution of visited states - standard i.i.d. generalization guarantees do not directly apply:

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Challenge: Distribution-specific guarantees

Even if we learn a model with low generalization error, the guarantees hold **only for the distribution** $\mathcal{D}$ from which the data was sampled.

When predictions are used for **decision-making** or **interaction with an environment**, this becomes critical:

- The data distribution during deployment may differ from $\mathcal{D}$
- There are **no guarantees** that performance will remain good under such shifts

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How does supervised learning relate to decision-making?

We will explore two key paradigms:

- **Behavior cloning (imitation learning):** learn a policy from expert demonstrations, treating actions as labels.
- **Active learning:** select which data to acquire.